

Six Months with AMD Strix Halo

Local AI, Open Source, and Hardware Reality

A Six Month Review of a Strix Halo Rig

linux | ROCm | Local AI | Home Assistant | Coding

winmutt | June 2026

github.com/winmutt

What We're Covering

Section 1: The Hardware Stack

- Power Consumption: Corsair 300 vs RTX 5090 vs Cloud
- The Hardware: AMD Ryzen AI Max+ 395 APU
- The Stack: ROCm + Lemonade Server
- Issue #1070: Core affinity across dual CCDs

Section 2: Projects & Applications

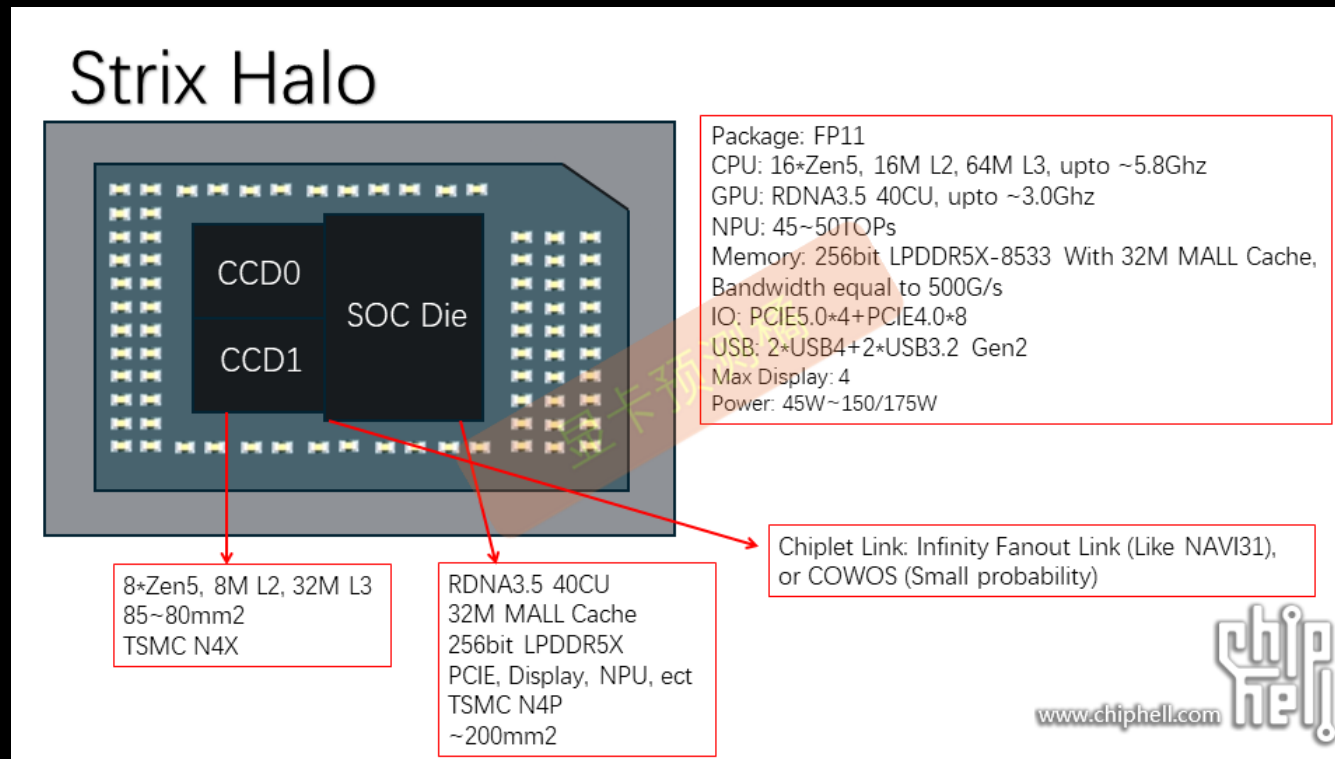
- Home Assistant: Cutting the Alexa cord
- 3D Modeling: AI to concrete signs
- AI Coding Editors: Opencode, Cline, Aider
- Project WWS: Remote workspace provisioning

Section 3: Open Source Journey

- Lessons Learned: What worked, what didn't
- Key Insights: 8 takeaways from 6 months
- OSS Contributions: 3.5x increase
- 26 Years: From LKML 2000 to Strix Halo

The Hardware: AMD Ryzen AI Max+ 395

APU Die Architecture



Physical Layout

- 16 cores (12 performance + 4 efficiency)
- 2x Core Complex Dies (CCD) with 32MB L3 cache each
- 768KB L1d + 512KB L1i + 16MB L2
- Integrated RDNA 3.5 GPU (40 Compute Units)
- 128GB LPDDR5X-8000 unified memory (soldered)
- Source: AMD Ryzen AI Max+ 395 teardown analysis (Chiphell, TechPowerUp)

AMD Radeon 680M GPU

- RDNA 3.5 architecture, 40 Compute Units
- Peak performance: ~4.6 TFLOPS
- Shared memory with CPU (128GB unified)
- Supports ROCm for AI/ML workloads
- Source: AMD Ryzen AI Max+ 395 specifications

AMD XDNA 2 NPU

- Neural Processing Unit for AI workloads
- Integrated into Ryzen AI Max+ 395 APU
- Supports FastFlowLM for efficient inference
- Power-efficient alternative to GPU processing

FastFlowLM Performance (June 2026):

- 6784 tokens prefill in ~15 seconds
- ~455 tokens/sec on NPU
- Chunked processing: 4096 + 2688 tokens
- End-to-end (prefill to response): ~16.2 seconds
- Source: FastFlowLM logs (lemonade-server)

Feat #1070: [enhancement] Core affinity when running multiple models.

Filed

February 8, 2026

Repository

github.com/lemonade-sdk/lemonade/issues/1070

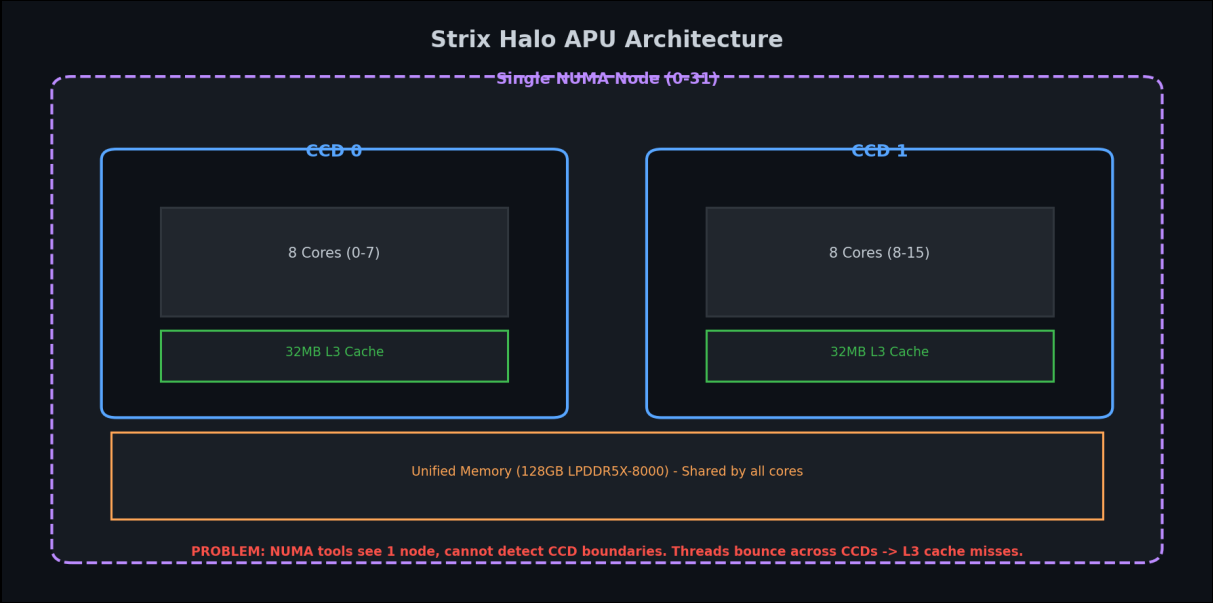
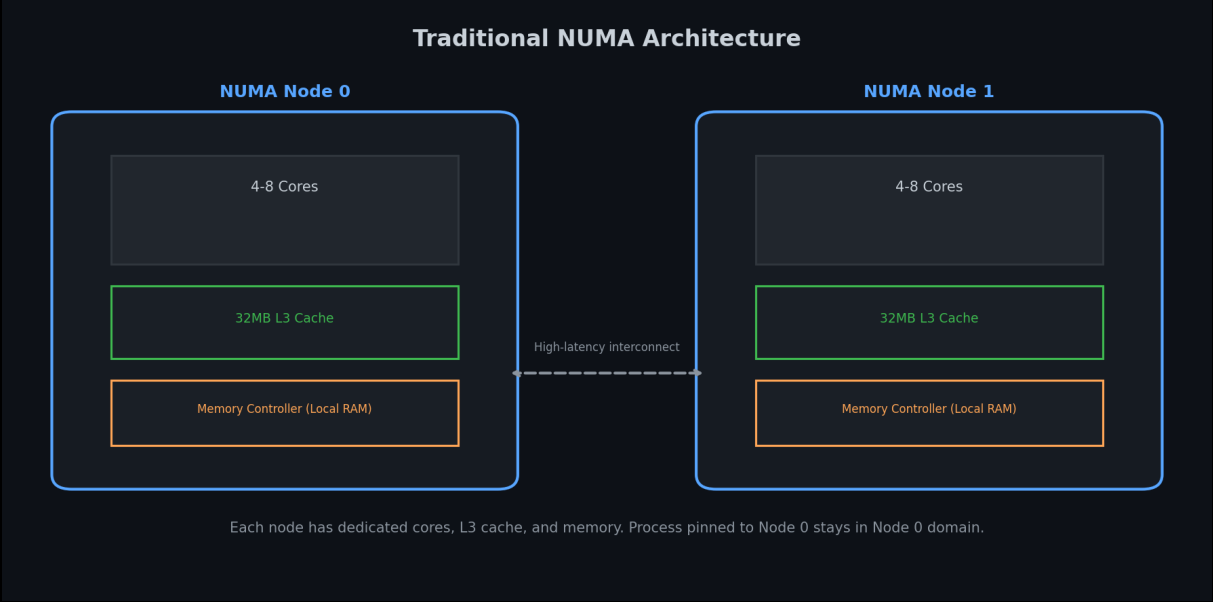
Status

Open

Hardware Reality

- One processor, but 2x core dies (CCDs) with separate 32MB L3 caches
- Traditional NUMA: 2 nodes, each with own cores + L3 cache
- Strix Halo: Single node, but 2 CCDs with separate L3 caches

Traditional NUMA vs. Strix Halo APU



The Problem: NUMA Tools Can't See CCD Boundaries

- Traditional NUMA tools (numactl, NUMATopologyFilter) detect only 1 node
- Cannot see CCD boundaries automatically
- Manual core pinning required to keep workloads on same CCD
- Running multiple LLMs: threads bounce across CCDs → L3 cache thrashing

Result: Cross-CCD traffic → slower inference, higher latency

The Solution (Proposed)

Manual core pinning (like Red Hat's vcpu_pin_set approach):

- Reserve cores per CCD, pin llama-server accordingly
- Pin based on number of models loaded and CCD boundaries
- Prevent cache-crossing penalties

Result: Each model stays in its own L3 cache domain → no cross-CCD traffic

Evidence: Threads Bouncing Across CCDs

ps Output: Threads Bouncing Across CCDs			
PID	LAST_CPU	AVG_MHZ	COMMAND
56840	6	55.7	llama-server (Model 1)
56840	20	54.2	llama-server (Model 1)
59798	0	51.2	llama-server (Model 2)
59798	2	47.5	llama-server (Model 2)

56840	22	55.7	llama-server (Model 1)
56840	4	54.2	llama-server (Model 1)
59798	16	51.2	llama-server (Model 2)
59798	18	47.5	llama-server (Model 2)

Notice: LAST_CPU jumps across CCD boundaries

CCD0: cores 0-7, 16-23 | CCD1: cores 8-15, 24-31

Hardware Topology: CCD Boundaries (lscpu)

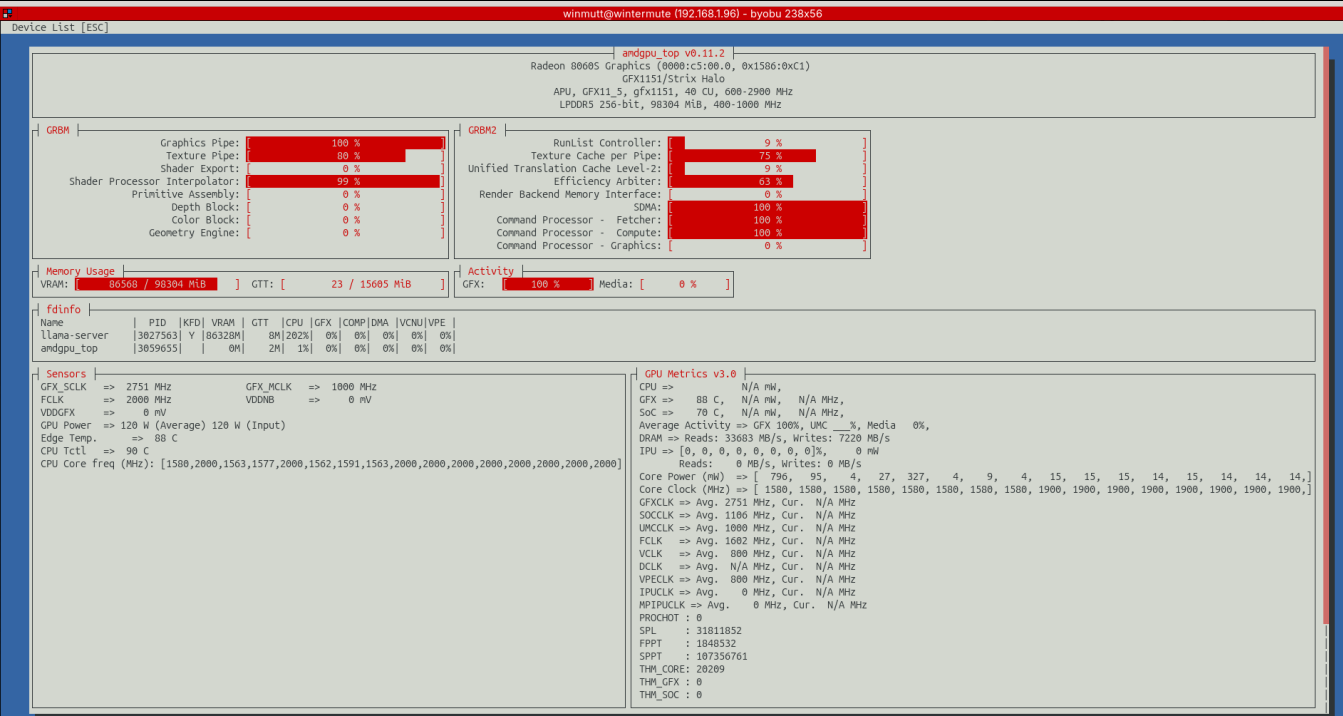
lscpu Output: CCD Topology					
CPU	NODE	CORE	L3	MAXMHZ	AVG_MHZ
0	0	0	0	5187.0	4937.8
1	0	1	0	5187.0	2000.0
2	0	2	0	5187.0	4909.0
3	0	3	0	5187.0	4904.9
4	0	4	0	5187.0	4889.9
5	0	5	0	5187.0	4845.2
6	0	6	0	5187.0	4882.0
7	0	7	0	5187.0	4950.4

8	0	8	1	5187.0	2000.0
9	0	9	1	5187.0	2000.0
10	0	10	1	5187.0	2000.0
11	0	11	1	5187.0	2425.1
12	0	12	1	5187.0	3056.8
13	0	13	1	5187.0	4661.1
14	0	14	1	5187.0	2000.0
15	0	15	1	5187.0	2000.0

L3 Cache + AVG_MHZ: Shows load shifting across CCD boundaries

NUMA tools see: 1 node (0-31) | Reality: 2 CCDs with separate L3 caches

Evidence: amdgpu_top



Cutting the Alexa Cord

LineageOS on Echo

December 2025

Started exploring XDA Forums for Echo device hacking

The Process:

1. Unlock bootloader via amonet exploit
github.com/R0rt1z2/amonet
2. Install TWRP recovery
3. Flash LineageOS 18.1 (Android 11)
4. Configure ViewAssist for Home Assistant
dinki.github.io/View-Assist

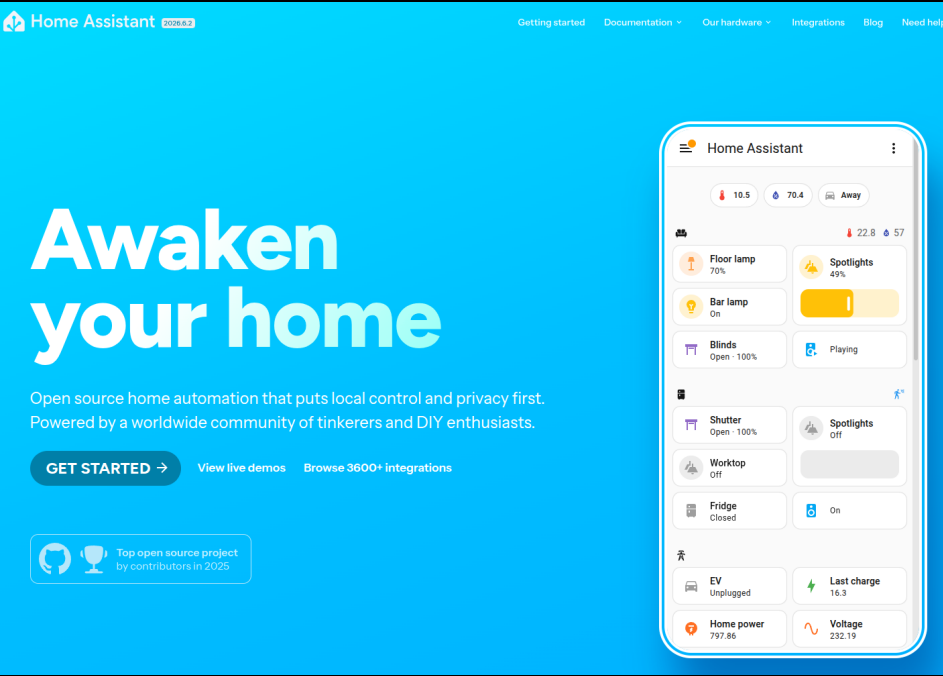
Philosophy

"Basically anything with a USB port. If there is a port, there is a way."

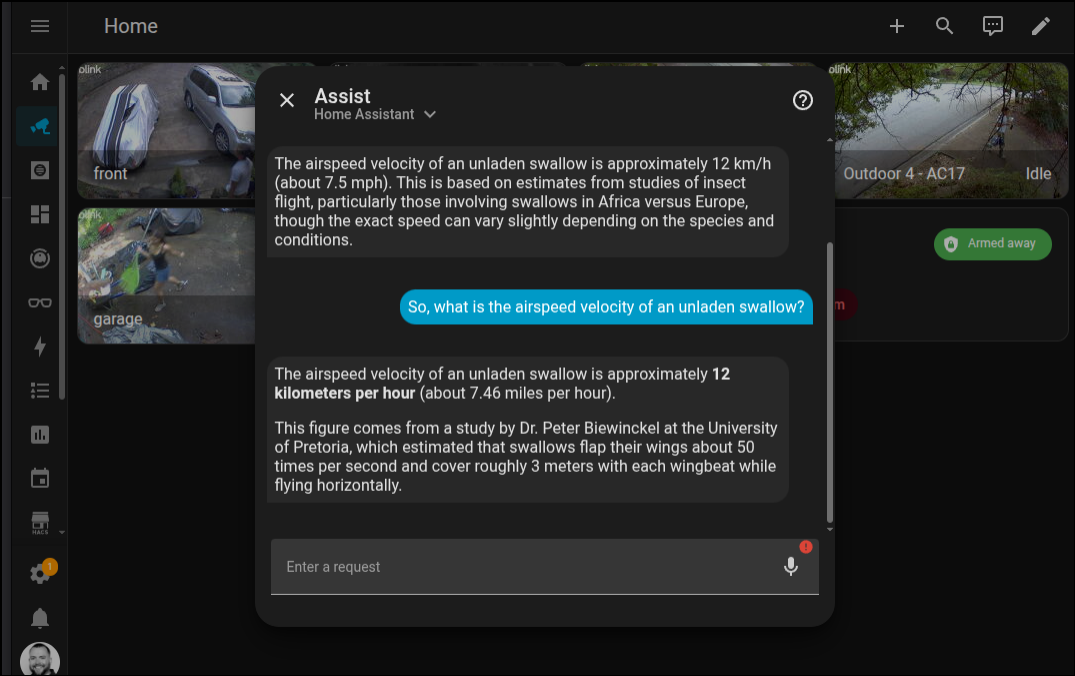
Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

Home Assistant Dashboard



Home Assistant on Echo Show



LineageOS Device Compatibility

Echo Show Devices Supported

Device	Codename	SoC	XDA Thread
Echo Show 5 (1st Gen 2019)	checkers	MT8163	xdaforums.com/t/rom-unofficial-11-checkers...
Echo Show 5 (2nd Gen 2021)	cronos	MT8163	xdaforums.com/t/rom-unofficial-11-cronos...
Echo Show 8 (1st Gen 2019)	crown	MT8163	xdaforums.com/t/rom-unofficial-11-crown...

All devices use MediaTek MT8163 SoC

LineageOS 18.1 (Android 11) - community builds by @R0rt1z2

Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed

January 7, 2026

Repository

<https://github.com/amazon-oss/releases/issues/4>

Status

Open (still waiting, it's fine)

The Problem

- Echo 5: Stable for days
- Echo 8: Dies after ~24 hours
- *Root cause: Unknown (yet)*

3D Modeling with Strix Halo

AI-Assisted Physical Design

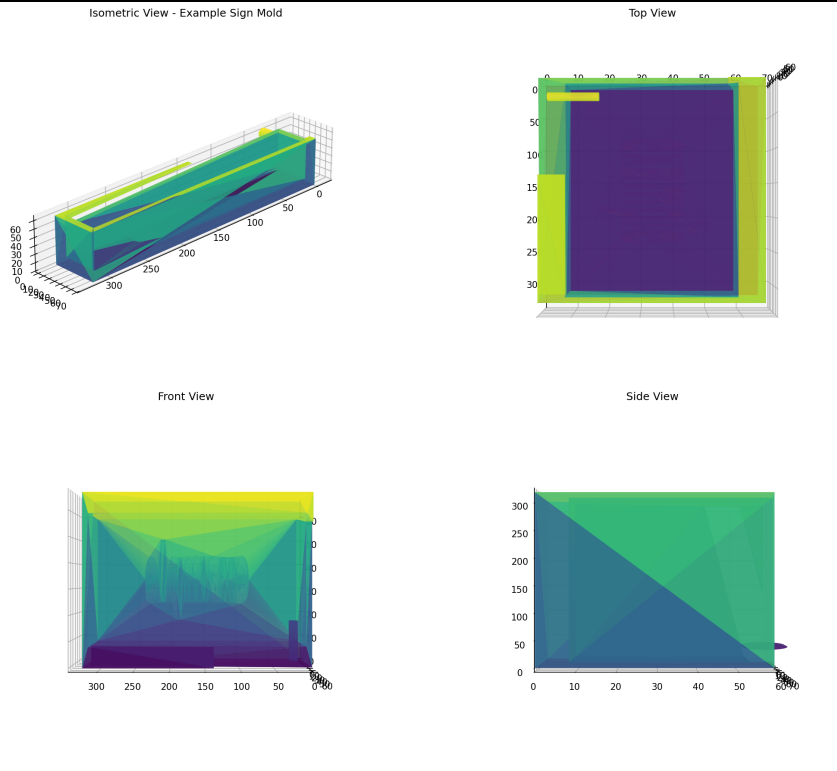
Blender + OpenSCAD workflow for creating physical objects

The Process

1. AI generates OpenSCAD/Blender design
2. Export to STL
3. 3D print mold
4. Create silicone mold
5. Cast concrete

December 1, 2025 — More physical projects coming soon

Concrete Sign Mold Design



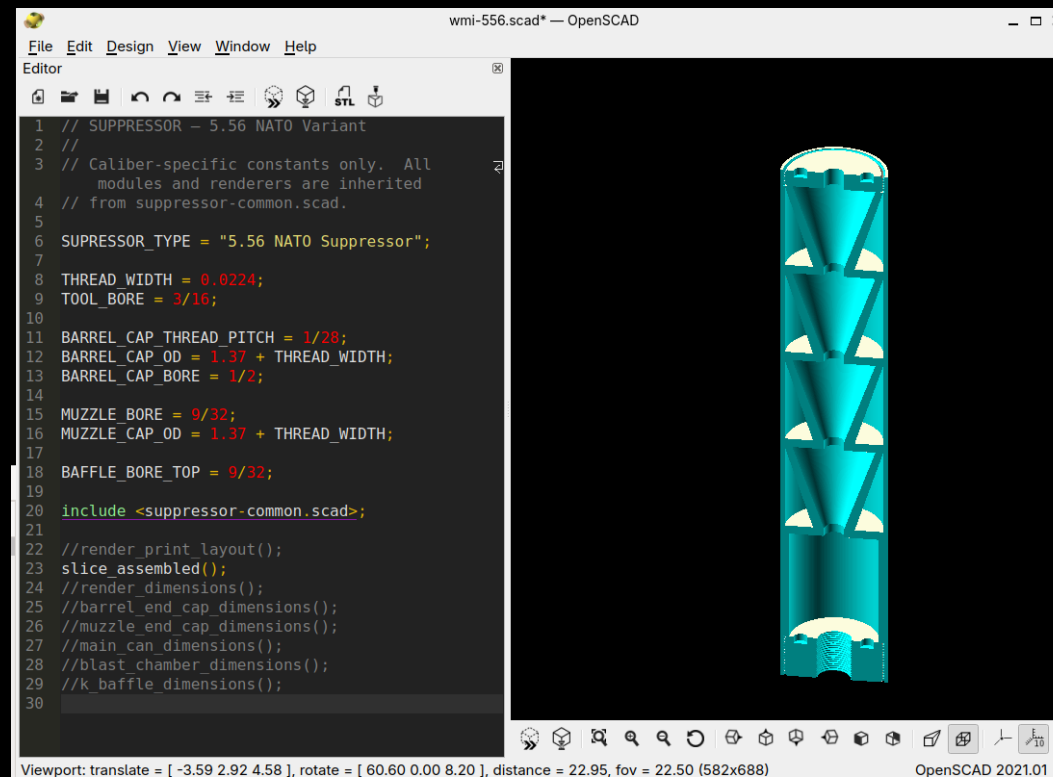
OpenSCAD: Where Things Work

wmi002 Session — Precision Engineering

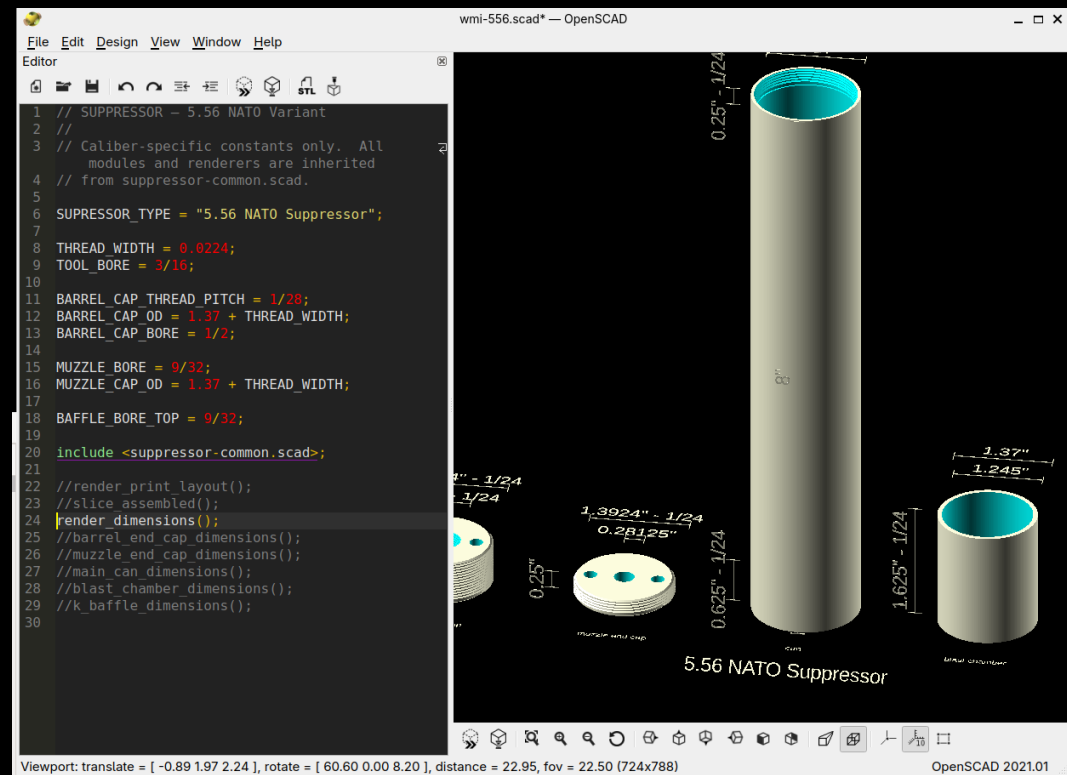
Success Story

- Parametric design: Caliber-specific constants (.308 Winchester, 5.56 NATO)
- Modular architecture: include, (BOSL2)
- Precision engineering: Thread pitches (5/8-24 UNF), bore dimensions, tolerances
- Working output: K-baffles, blast chambers, threaded end caps

K-Baffle Assembly (Internal View)



Dimensioned Parts View



Two Approaches to Physical Design

***Lesson:** OpenSCAD sessions produce working parts. Blender sessions produce concrete. Both are valid. Only one has thread tolerances.*

Project WWS: I Vibe-Coded an Entire System

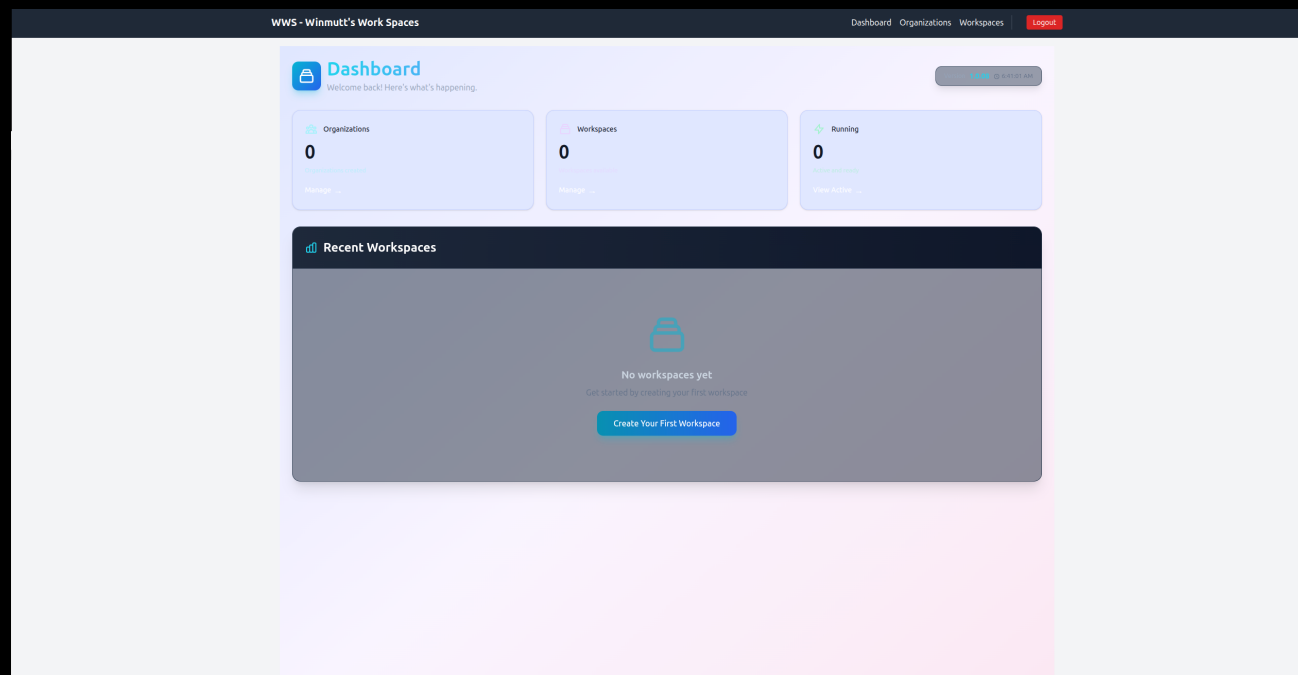
Winmutt Work Spaces — github.com/winmutt/www

May 12, 2026

"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."

Remote workspace provisioning with KVM/Podman isolation
code-server (VSCode in browser) + GitHub OAuth + RBAC

First commit: Feb 22, 2026 → Production: May 12, 2026



Lessons Learned (Mostly)

What Worked ✓

- Cline + Qwen3-Coder: Actually produces code
- Prompt Caching: Things hum now
- AMD GPU Libraries: 2x faster (worth the pain)
- WWS Project: Vibe-coded from Feb 22 (first commit) to May 12
- Home Assistant: Alexa is dead, long live OSS

What Didn't ✗

- Memory: 32GB reserved for Lemonade/llama.cpp/opcode
- Ollama: Still crashes after updates
- NPU: Coming soon™ (always)
- Context >64k: TPS drops like a stone

1. Hardware hype $\neq$ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI $\rightarrow$ physical objects (concrete signs, apparently)
6. Echo $\rightarrow$ Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

26 Years of Open Source (2000-2026)

From LKML to Strix Halo: How Hardware Bought My Time

The Long Detour

For 20+ years I built corporate SaaS products. Cloud disconnected me from the metal. Linux became something I deployed to, not something I touched daily. The kernel, the drivers, the hardware—left that world behind. Then Strix Halo arrived and changed everything.

June 9, 2000: Asking on LKML

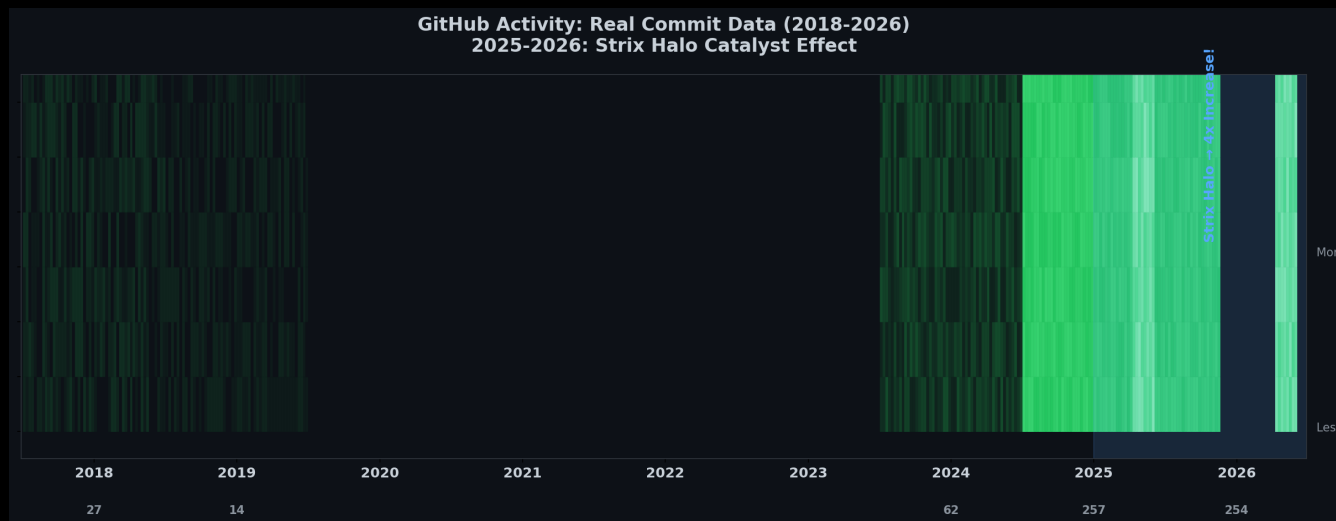
From: Rolf Martin-Hoster (winmutt@hotmail.com)
Date: Thu Jun 08 2000 - 19:24:59 EST
Subject: HighPoint IDE RAID (kernel 2.2.14)

Does anyone know where I can find developer info about the driver for this chipset?

-Rolf

Back then: Hands-on with kernel 2.2.14, digging into IDE RAID drivers, asking questions on linux-kernel mailing list.

The Return: GitHub Activity (2018-2026)



2018-2024: Ghost town. 2025: Strix Halo arrives. November 2025: Back on the metal, every single day. AI + local hardware reinvigorated my love for open source.

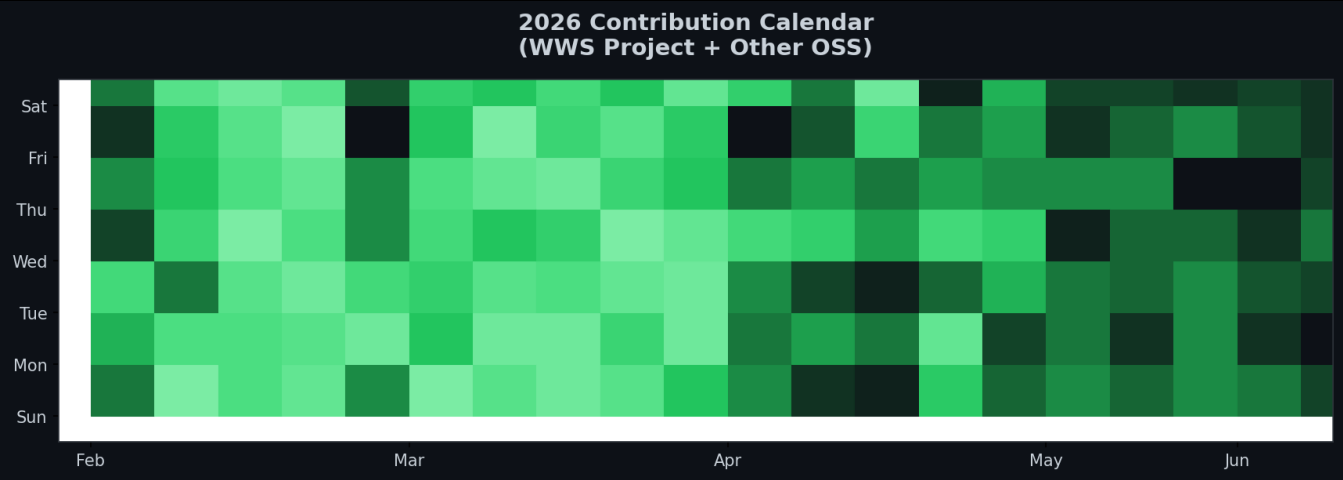
Moral: Sometimes the best way back in is to buy new toys.

How Buying Hardware Got Me Back in Open Source

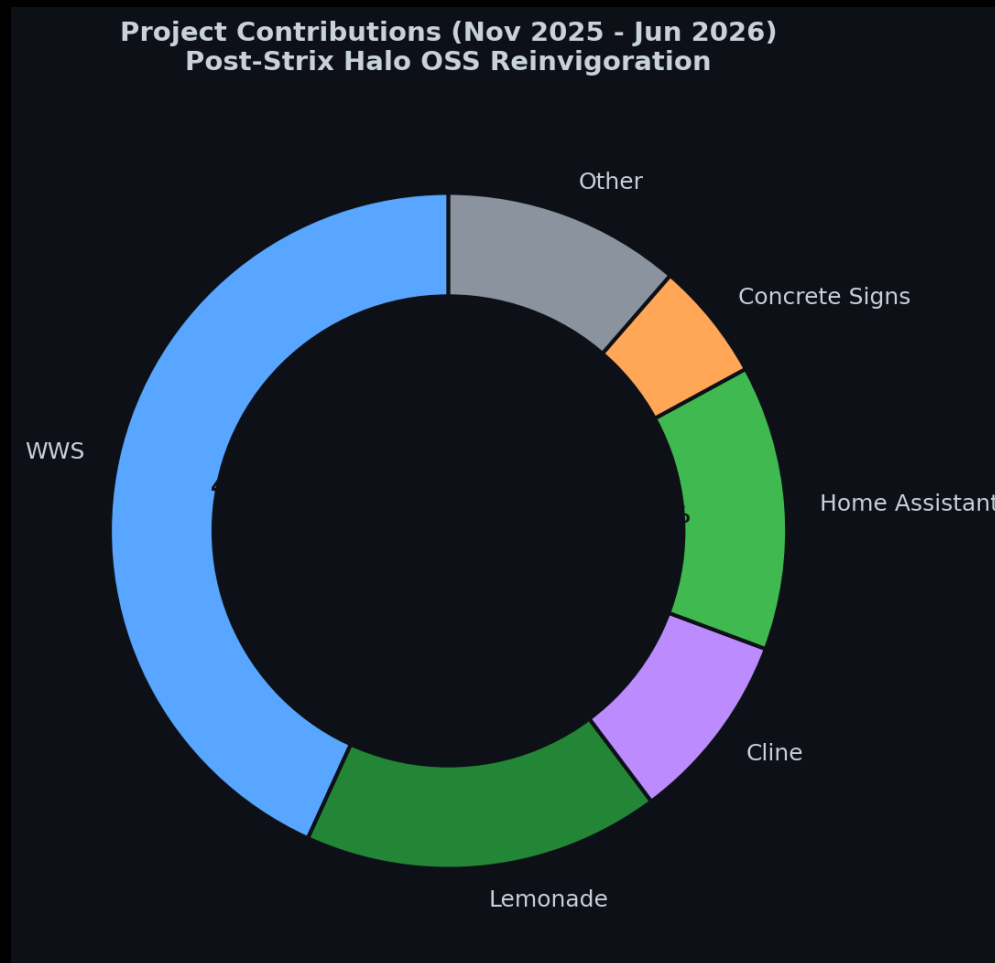
The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

2026: The Year I Came Back



Projects I've Actually Contributed To



Contribution Breakdown

- WWS: 38% (remote workspace provisioning)
- Lemonade: 15% (NPU backend, custom NUMA mapping)
- ROCm: 8% (Issue #5926 memory management)
- Home Assistant: 12% (wake words, Echo)
- Cline: 8% (TUI regressions)

- Concrete Signs: 5% (Blender/OpenSCAD)
- Other: 10% (miscellaneous)

AI Coding Editors: My Daily Drivers

Opencode

Most usable, portable, and stable. Best agentic coding experience. Runs as a web server (<http://localhost:3000>) allowing remote access with a localized agent running for autonomous development. github.com/anomalyco/opencode

Cline

Good TUI but many regressions over time.

Also Use: Koo Roo, Aider

References & Sources

Power Consumption Sources

- AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/
- NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
<https://www.nvidia.com/geforce/rtx-5090>
- NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
<https://www.nvidia.com/geforce/rtx-4090>
- Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
<https://www.corsair.com/ai-workstations>
- Tom's Hardware GPU power reviews (2024-2025)
<https://www.tomshardware.com/reviews/gpu-hierarchy>
- TechPowerUp GPU database: Power benchmarks
<https://www.techpowerup.com/gpu-specs>
- NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

- Lemonade: AMD's llama.cpp + ROCm backend (github.com/winmutt/lemonade)
- Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)
- ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)
- ROCm Issue #5926: Memory management bugs (github.com/ROCm/ROCm/issues/5926)
- WWS: github.com/winmutt/wws
- Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection
- NUMA/CPU Pinning: Red Hat OpenStack docs ([vcpu_pin_set](#), [NUMATopologyFilter](#))

https://docs.redhat.com/en/documentation/red_hat_openshift_platform/10/html/instances_and_images_guide/ch-cpu_pinning

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

Questions? (I Have Answers, Mostly)

1. Is Strix Halo production-ready? (It's complicated)
2. When will NPU support mature? (Coming soon™)
3. How do we solve 128GB addressing? (We don't)
4. Best backend for multi-model? (Lemonade, but...)
5. Is autonomous dev the future? (Ask my AI)

github.com/winmutt

Come find me after - I'll show you the rig